ICT330

Timed Online Assignment – July Semester 2021

Database Management Systems

Monday, 15 November 2021 **1:00 pm – 3:30 pm**

Time allowed: 2.5 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **FOUR (4)** questions and comprises **SEVEN (7)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, Your Name, and SUSS PI No. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT Space. *Example: TOAICT330_RaphaelLee_T1923161 (omit D/O, S/O).*

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a Microsoft Word file saved in .doc or .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the Word file provided it does not exceed the file size limit of 500MB.

7. You are required to sketch a diagram in some of the questions, if you are unable to include them into the word document, you may upload them as pdf.

   Please ensure that the total file size including word document does not exceed 500 MB and rename your pdf with the following naming convention: TOACourseCode_FullName_StudentPI_PDF1

   All uploaded hand written sketches and diagrams must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

Answer all questions. (Total 100 marks)

Question 1

You are developing a website for virtual tours. You have gathered the necessary data requirements for the application.

A virtual tour will have a tour id and is conducted in one country. One or more places in the country can be included in a virtual tour, but places are unique to tours. The description of the places and their names are to be listed together with a tour.

Each tour will be conducted by one or more guides, and each guide is qualified to conduct at least one tour. The guide(s) conducting a virtual tour on a particular tour date must be recorded. For some places e.g., zoo, employed staff of the place will act as tour guides. For such staff, it is important to record when the staff start conducting tours. For other places e.g., wetland, freelancers act as tour guide. Therefore, it is important to include a contact number. All guides are identified by an id and personal details such as name, date of birth and gender.

Review and rating are solicited after a virtual tour is conducted. Although the reviews and ratings are for a specific conducted virtual tour, they are listed with the virtual tour in general, and further, they can be anonymous or may include a customer name.

Your application keeps track of places that are not yet in any virtual tour so that they can be incorporated in future virtual tours.

Construct **ONE (1)** conceptual (ER) model from the statements of requirements to represent the data model, showing both

(a) entities, with entities name, relevant attributes and identifier,

(10 marks)

(b) relationships with maximum and minimum cardinalities and relationship name.

(10 marks)

State your assumption(s) for only data requirements that are not specified.

Question 2

An agency can list one or more jobs in a job placement portal used by several agencies. Each job is assigned a unique job number and has one or more job categories e.g., Research and Analysis, Social and Community Development, Statistics. Jobs may have overlapping categories. Each category has a category code.

Each job is for a specific position and has an application closing date. The required education qualification is also indicated. Different agencies have different minimum salary for different education qualification.

You are given the Job table:
Job (agency, jobNumber, position, closingDate, requiredQualification, minSalary, category, categoryCode)

State your assumption(s) for only data requirements that are not specified.

(a) Formulate and list the functional and multi-valued dependencies.

(6 marks)

(b) Propose candidate key(s) for the table.

(2 marks)

(c)

(i) For each MVD in answer to part a), state whether it is subsumed.

(ii) Normalise the table to BCNF and 4NF by applying **the BCNF approach** covered in the course text.

Show how you apply the steps to arrive at the tables, and explain whether each derived table at each iteration is already in BCNF and 4NF (if applicable).

(10 marks)

(d) Assemble your BCNF and 4NF tables and constraints clearly by giving each table and the referential integrity constraint(s), if any, in the forms described below:

Tables in this form:
TABLE-NAME(col-name$_1$, col-name$_2$, ..., col-name$_n$)

- indicate (PK) against the primary key and
- indicate (FK) against the foreign keys, if any.

Referential integrity constraint(s), if any, **together with the child table** in this form:
TABLE-NAME$_x$.col-name$_{x1}$ ...col-name$_{xn}$ must exist in
TABLE-NAME$_y$.col-name$_{y1}$...col-name$_{yn}$

(7 marks)

Question 3

(a)

(i) Transform the ERD in Figure Q3(a)(i) into a logical model.

Figure Q3(a)(i)

(ii) Refer to the description of the columns for the column definition, and then write the appropriate SQL statements to implement only these two tables: LiveVirtualTour and Booking, derived from the ERD in Figure Q3(a)(i).

List your SQL statements in the order that they should be executed.

Description for columns in the tables:

- LiveVirtualTour
  - All columns are required,
  - tourCode is 5 characters long, it starts with either the letter A or L followed by 4 digits.
  - name is at most 30 characters long.
  - tourType 11 characters long, and is either cultural, cooking or sightseeing.
  - cost is a decimal number,
  - capacity is a whole number from 1 to 10, inclusive of 1 and 10.
- Booking
  - All columns are required,
  - bookingId is auto-generated, it starts from 1 with increment of 1.
  - bookedForTourDate is date that is at least 3 days away from the date the booking is made, assumed to be today.
  - numPax is a whole number.
- Customer
  - All columns are required,
  - custEmail is at most 40 characters long.
  - name is at most 30 characters long.

- Constraint

  C1: Minimum cardinality between LiveVirtualTour and Booking for the <u>isFor</u> relationship is 1:N, M:O with delete and update disallowed.

  C2: Minimum cardinality between Customer and Booking for the <u>makes</u> relationship is 1:N, M:O with delete and update cascaded.

(10 marks)

(b) Formulate SQL queries to display the required result sets:

(i) Display the details of tours with capacity more than 5 and that cost less than $50. Show the tours in ascending order of cost and descending order of capacity.

(ii) Display the details of cultural tours booked for this year by customer dennis@gmail.com. Use subquery.

(iii) Display the number of times each of the various types of tours (cultural, cooking and sightseeing) have been booked. If a type of tour has not been booked, display 0 for the number of times booked for that type of tour. Do not hardcode the types of tours, i.e., do not hardcode cultural, cooking and sightseeing.

(15 marks)

(c) Write a trigger to enforce the constraints:

- A live virtual tour cannot be booked more than once for a same tour date. If it is booked more than once for a same date, display the error message: Same date
- numPax in any booking cannot be more than the capacity of the live virtual tour the booking is for. If the capacity is exceeded, display the error message: Capacity exceeded
- If the constraints are not violated, display the message: Successful operation

State any assumptions you made in implementing the trigger.

(10 marks)

Question 4

(a) Demonstrate the basic techniques of concurrency control and their application. Consider a web-based application for live virtual tours and the use cases: make booking and update booking.

```
1. begin transaction
2. insert into Booking
   values (---)
3. commit
```

Make Booking

```
1. begin transaction
2. update  Booking
   set  bookedForTourDate = …
   where bookingId = …
3. commit
```

Update Booking

Suppose all three tables described in Q3 have been implemented including the trigger described in Q3c).

Discuss and explain whether setting the isolation level to the 4 types of isolation levels (read uncommitted, read committed, repeatable read, serializable) can cause deadlocks, lost updates and violation of the constraint that dates in the column bookedForTourDate must be unique for the same tour. State any assumptions you make.

(8 marks)

(b) Consider the booking system described in part a). Describe a permission that should be given to customers and a permission that should not be given. Explain why they should or should not be given to customers.

(4 marks)

(c) Compare and contrast the database recovery procedure actions for these pair of transactions when a crash happens after DBMS has serviced all their requests, and explain why the DBMS must undertake the recovery actions in your answer:

(i) A transaction that rollbacks before a checkpoint and a transaction that rollbacks after a checkpoint.

(4 marks)

(ii) A transaction that commits after the checkpoint and a transaction that is still ongoing after the checkpoint.

(4 marks)

----- END OF PAPER -----